

Integration — Lesson 36 Test: Arc Length

10 questions · show your work in the box · formula: $L = \int_a^b \sqrt{1 + (f'(x))^2} dx$ · exact answers only (fractions and radicals), no decimals · "set up" means write the integral and STOP

Name: _____ Date: _____ Score: _____ / 10

1. Use the arc length formula on $f(x) = (4/3)x$ over $[0, 9]$. Then verify your answer with the distance formula between the two endpoints. 1 pt

2. Find the exact length of $f(x) = 2x - 5$ over $[1, 4]$. (The answer is not a whole number — leave it exact, and check it with the distance formula.) 1 pt

3. Find the length of $f(x) = (2/3)x^{3/2}$ over $[0, 15]$. Show the simplification of $1 + (f')^2$ before you integrate. 1 pt

4. Does adding a constant change a curve's length? Find the length of $f(x) = (2/3)x^{3/2} + 11$ over $[0, 24]$, and explain in one sentence what the +11 does (or does not do) to the answer. 1 pt

5. Write — but do not evaluate — an integral for the length of $y = x^2$ over $[1, 3]$. 1 pt

6. Write — but do not evaluate — an integral for the length of $y = \sin x$ over $[0, \pi/2]$, and say in one sentence why we stop at the setup. 1 pt

7. Set up (do not evaluate) an integral for the length of each curve: (a) $y = e^x$ over $[1, 2]$; (b) $y = \ln x$ over $[1, 4]$. 1 pt

8. A curve has derivative $f'(x) = \sqrt{4x^2 + 4x}$. Simplify $1 + (f')^2$ until the square root disappears, then find the curve's length over $[0, 3]$. 1 pt

9. Asked for the length of $f(x) = (5/12)x$ on $[0, 12]$, a student writes $L = \int_0^{12} \sqrt{(f')^2} dx = \int_0^{12} (5/12) dx = 5$. (a) Name the error. (b) Say what quantity the student actually computed. (c) Give one quick reason 5 is impossible here. (d) Give the correct length. 1 pt

10. Find the exact length of $f(x) = x^{3/2}$ over $[0, 5]$. Then compute the straight-line distance between the endpoints and confirm your arc length is longer, as it must be. 1 pt
